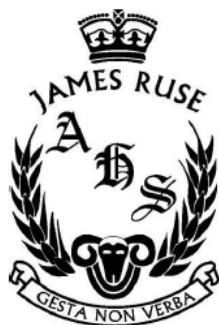


Name: _____

Class: _____



2018

Higher School Certificate

Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7–10)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I Write your answers on the multiple choice answer sheet provided

1. It is given that $\log_a 8 = 1.893$, correct to 3 decimal places.
What is the value of $\log_a 4$, correct to 2 decimal places?

(A) 0.95
(B) 1.26
(C) 1.53
(D) 2.84

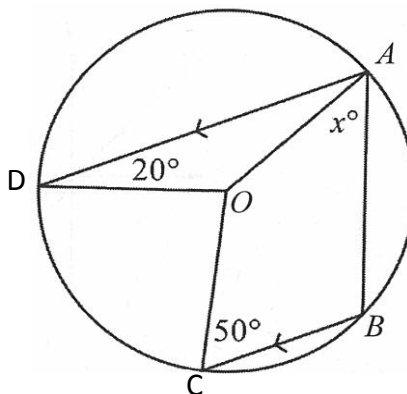
2. Which group of three numbers could be the roots of the polynomial equation

$$x^3 + px^2 - 26x + 24 = 0 ?$$

(A) 1, 3, 4
(B) -1, -3, -8
(C) -1, -2, 12
(D) 1, -6, 4

3. A, B, C and D are points on a circle with centre O. BC is parallel to AD.
 $\angle ADO = 20^\circ$ and $\angle BCO = 50^\circ$.
Let $\angle BAO = x^\circ$

Not to scale



What is the value of x ?

(A) 15
(B) 35
(C) 40
(D) 55

4. Which expression is equal to $\int \cos^2 \frac{2x}{5} dx$?

(A) $\frac{x}{2} + \frac{5}{8} \sin \frac{4x}{5} + c$

(B) $\frac{x}{2} - \frac{5}{8} \sin \frac{4x}{5} + c$

(C) $\frac{x}{2} + \frac{2}{5} \sin \frac{4x}{5} + c$

(D) $\frac{x}{2} - \frac{2}{5} \sin \frac{4x}{5} + c$

5. What is the value of $\lim_{x \rightarrow \infty} \frac{3x+7}{2-x}$?

(A) 3

(B) -3

(C) $\frac{7}{2}$

(D) $-\frac{7}{2}$

6. A particle is moving in simple harmonic motion along a straight line according to the equation $v^2 = -x^2 + 2x + 8$, where v is its velocity and x its displacement. What is the amplitude of the motion?

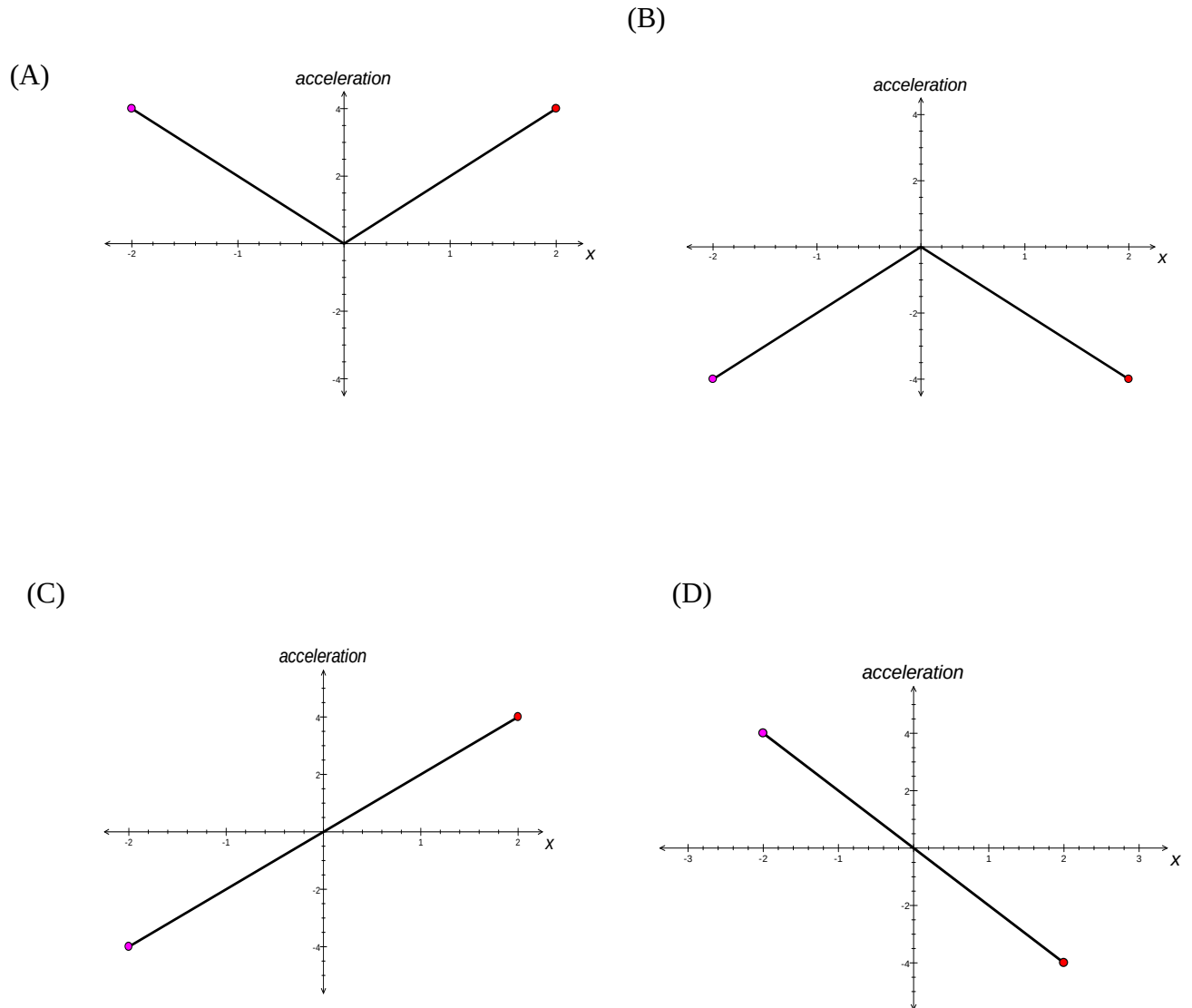
(A) 2π metres

(B) 3 metres

(C) 8 metres

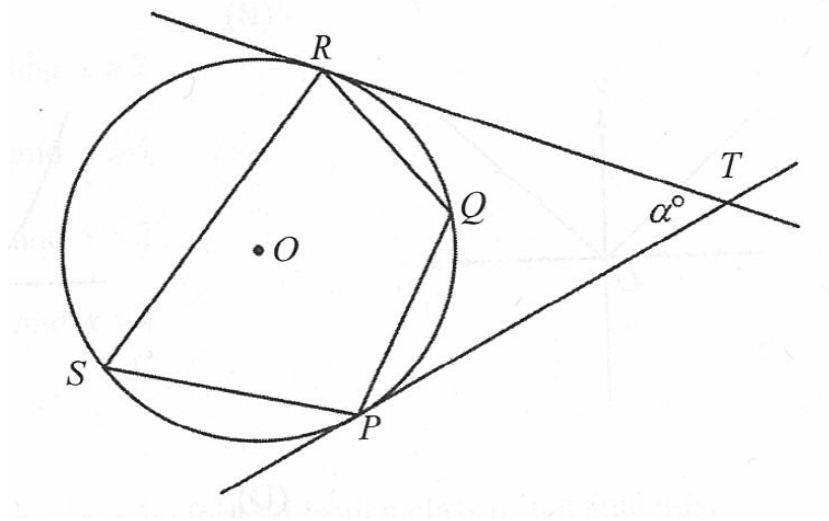
(D) 9 metres

7. A particle is moving along a straight line. The displacement of the particle from a fixed point O is given by x . The graphs below show acceleration against displacement. Which of the graphs below best represents a particle moving in simple harmonic motion?



8. The points P, Q, R and S lie on a circle with centre at O.

The tangents at P and R meet at the point T and $\angle RTP = \alpha^\circ$



What is the size of $\angle PQR$ in terms of α ?

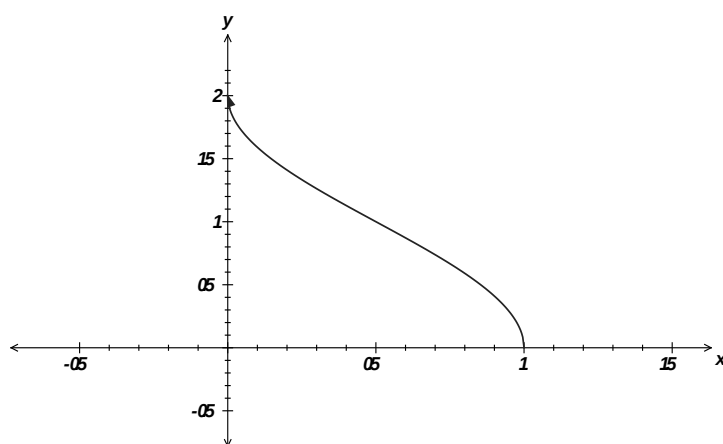
- (A) $(180 - \frac{\alpha}{2})^\circ$
- (B) $(180 - \alpha)^\circ$
- (C) $(90 + \frac{\alpha}{2})^\circ$
- (D) $(90 + \alpha)^\circ$

9. When a dishwasher has completed its cycle, and the contents removed, metallic utensils cool such that the temperature of the utensils is given by $T = 20 + Ae^{-kt}$ where t is the time in minutes after the utensils have been removed from the dishwasher. The utensils cool from 50° to 35° in 5 minutes.

The exact value of k is?

- (A) $\frac{1}{5} \log_e 2$
- (B) $\frac{1}{5} \log_e \frac{1}{2}$
- (C) $\frac{1}{5} \log_e \frac{10}{3}$
- (D) $\frac{1}{5} \log_e \frac{3}{10}$

10.



Which of the following could be the equation of the function shown above?

(A) $y = 2 \cos^{-1}(2x - 1)$

(B) $y = 2 \cos^{-1}\left(\frac{x-1}{2}\right)$

(C) $y = \frac{2}{\pi} \cos^{-1}(2x - 1)$

(D) $y = \frac{2}{\pi} \cos^{-1}\left(\frac{x-1}{2}\right)$

Section II begins on the next page

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Start each question on a new page. Extra paper is available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/ or calculations.

Question 11 (15 marks) Start a new page.

(a) Differentiate $e^{2x} \log_e 3x$ **2**

(b) Find the primitive function of $\frac{1}{1+9x^2}$ **2**

(c) Solve $\frac{x^2+3}{x} > 4$. **2**

(d) The equation $\cos 2x = e^{-x}$ has a root near $x = 0.4$
Taking $x = 0.4$ as a first approximation, use Newton's method to find a second approximation to the root of the equation.
Give your answer correct to two significant figures. **2**

(e) Given that $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$, use an appropriate expansion of $\cos 2\theta$ to find the exact value of $\cos 36^\circ$. **3**

(f) A particle is moving such that its displacement (x metres) from a fixed point O after t seconds is given by $x = 3 - 2 \cos 2t$.

(i) Show that the particle is moving in simple harmonic motion. **2**

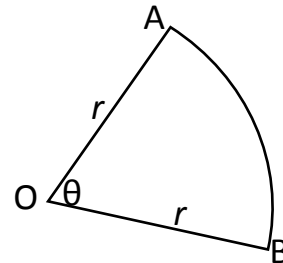
(ii) Find the maximum speed of the particle. **2**

Question 12 (15 marks) Start a new page.

- (a) The general solution of the equation $\sin 3x = -\frac{1}{2}$ has the form $x = nA + (-1)^n B$, where n is an integer. Write down a value for A and B . 2

- (b) AOB is a sector of a circle with centre O and radius r cm, as shown in the diagram.

$\angle AOB = \theta$ radians



- (i) The area of the sector AOB is 100 cm^2 . Show that $\theta = \frac{200}{r^2}$. 1
- (iii) If the radius is increasing at a constant rate of 0.5 cm s^{-1} , find the rate at which $\angle AOB$ is decreasing when $r = 10 \text{ cm}$. 2
- (c) Find the coefficient of x^3 in the expansion of $(3x - \frac{2}{x})^9$ 3
- (d) Thu plays Garry in a game in which, in any one game, Thu's chances of winning is 0.6 and drawn games are not possible.
- (i) If they agree to play 5 games, find the probability that Thu wins by 3 games to 2. (answer to 3 decimal places) 2
- (ii) If they agree to play until one of them wins 3 games, find the probability that Thu wins by 3 games to 2. (leave answer in unexpanded form) 2
- (e) A rocket is fired vertically upwards from the surface of the earth at a velocity of 5 km s^{-1} . The acceleration of the rocket is given by $\ddot{x} = -\frac{80000}{x^2} \text{ km s}^{-2}$, where x is the distance of the rocket from the centre of the Earth. The radius of the Earth is 6400 km
- (i) Show that the rocket never comes to rest. 2
- (ii) Find the rocket's velocity when it is 3600 km above the Earth's surface. 1

Question 13 (15 marks) Start a new page.

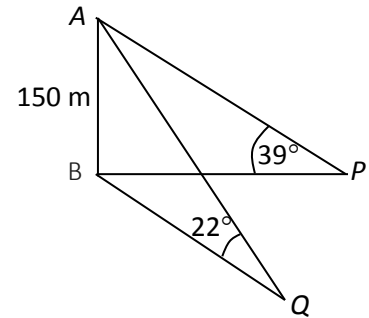
- (a) A is 150 metres above the horizontal plane BPQ .

3

AB is vertical. The angle of elevation of A from P is 39° and the angle of elevation of A from Q is 22° .

P is due East of B and Q is on a bearing of 138°T from B .

Calculate the distance from P to Q , to the nearest 10 metres.



- (b) Find the volume of the solid formed when the region enclosed between the curves $y = \sin x$ and $y = \sin 2x$ over the domain $0 \leq x \leq \frac{\pi}{3}$ is rotated about the x axis

3

- (c) Use mathematical induction to prove that

$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \frac{1}{3 \times 4 \times 5} + \dots + \frac{1}{n(n+1)(n+2)} = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$$

for all integers $n \geq 1$.

3

- (d) Four couples, each consisting of a male and a female, sit around a circular table. In how many different ways is it possible to seat the four couples so that males and females sit in alternating positions and nobody sits next to his or her partner?

2

- (e) A rocket is fired from a pontoon on the sea. The rocket is aimed to land on a flat 60m high cliff which is 200 m from the pontoon. The angle of projection of the rocket is 45° and its initial velocity is $40\sqrt{2}$ m/s.

- (i) Taking the point of projection as the origin O , derive expressions for the horizontal component $x(t)$ and the vertical component $y(t)$ of the position of the rocket at time t seconds. (Assume the acceleration due to gravity is 10m/s^2 and neglect air resistance).

2

- (ii) Find the distance the rocket lands from the edge of the cliff.

2

Question 14 (15 marks) Start a new page.

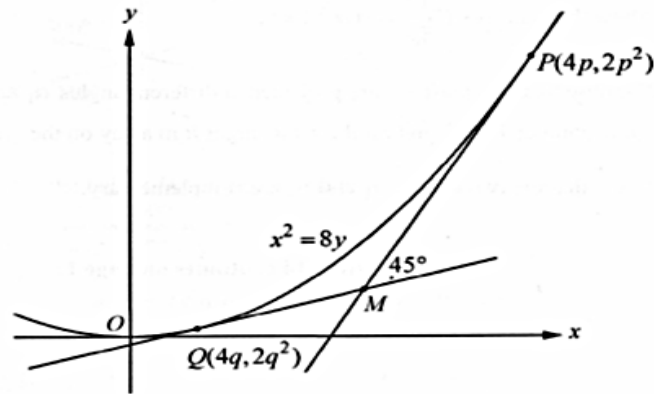
(a) Let $f(x) = \frac{1}{\sqrt{1+x^2}}$ for $x \leq 0$.

Find an expression for the inverse function $f^{-1}(x)$ in terms of x

2

(b) The points $P(4p, 2p^2)$ and $Q(4q, 2q^2)$ lie on the parabola $x^2 = 8y$.

The tangents at P and Q intersect at M .



The acute angle between these tangents is 45° .

(i) Find the coordinates of M .

2

(ii) Show that $|p - q| = |1 + pq|$

1

(iii) By eliminating the parameters, show that the equation of the locus of M is

$$x^2 - y^2 - 12y - 4 = 0$$

2

(c) Use the substitution $u = t^2 + 2$ to find $\int t^3 \sqrt{t^2 + 2} dt$

2

(d) Let $f(x) = 2 \cos^{-1} x$ $-1 \leq x \leq 1$
and $g(x) = \sin^{-1}(2x^2 - 1)$ $-1 \leq x \leq 0$

(i) Sketch the graph of $y = f(x)$

1

(ii) Show that $g'(x) = f'(x)$, $-1 < x < 0$

2

(iii) Hence, or otherwise, express $g(x)$ in terms of $f(x)$

2

(iv) Sketch the graph of $y = g(x)$

1

END OF EXAMINATION

Multiple Choice

① $\log_a 8 = 1.893$

$$\log_a 2^3 = 3 \log_a 2 = 1.893$$

$$\log_a 2 = \frac{1.893}{3}$$

$$\log_a 2^2 = 2 \times \frac{1.893}{3} = 1.262$$

B.

② $\alpha + \beta + \gamma = -p$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -26$$

$$-6 - 24 + 4 = -26 \checkmark$$

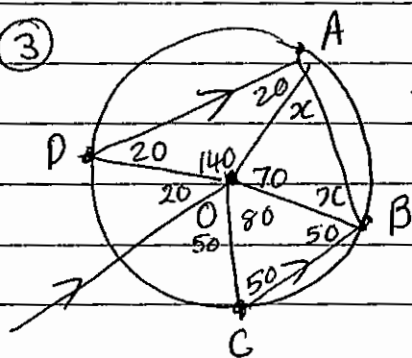
$$3 + 8 + 24 \neq -26$$

$$\gamma\beta\alpha = -24$$

only B or D

D

③



Construct OF || AD

Join O to B

$$\angle DOC = 20 + 50 = 70$$

$$\angle DOA = 180 - 20 - 20 = 140$$

$$\angle COB = 180 - 50 - 50 = 80$$

$$\angle BOA = 360 - 80 - 70 - 140 = 70$$

$$2x + 70 = 180$$

$$2x = 110$$

$$x = 55$$

D

④ $\int \cos^2 \frac{2x}{5} dx = \int \frac{1}{2} (1 + \cos \frac{4x}{5}) dx$

$$\cos^2 A = \frac{\cos^2 A - \sin^2 A}{2}$$

$$= \frac{1}{2} \left(x + \frac{5}{4} \sin \frac{4x}{5} \right) + C$$

$$= 2\cos^2 A - 1$$

$$= \frac{x}{2} + \frac{5}{8} \sin \frac{4x}{5} + C$$

$$\cos^2 A = \frac{1}{2} (1 + \cos 2A)$$

$$\cos^2 \frac{2x}{5} = \frac{1}{2} (1 + \cos \frac{4x}{5})$$

A

⑤ $\lim_{x \rightarrow \infty} \frac{3x+7}{2-x}$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3x}{x} + \frac{7}{x}}{\frac{2}{x} - \frac{x}{x}}$$

$$= \frac{3}{-1} = -3$$

B

⑥ $v^2 = -x^2 + 2x + 8$

At endpoints $v=0$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$\therefore$ Endpoints are $x = -2$ & $x = 4$

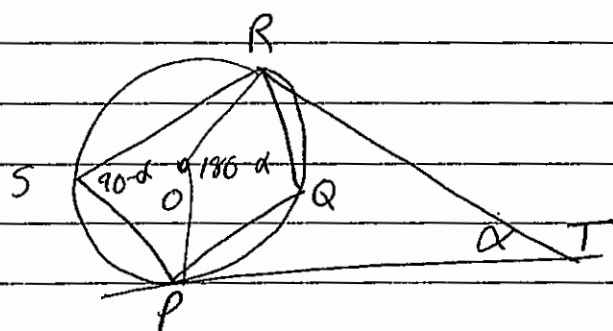
Centre is at $x=1$

$\therefore$ Amplitude = 3

A B

⑦ D

⑧ $\angle TRO = \angle PPO = 90^\circ$ (tangent perp to radius at pt of contact).



$\angle ROP + \alpha = 180$ (opp $\angle$'s cyc quad)

$$\angle ROP = 180 - \alpha$$

$$\angle ROP = 180 - \alpha$$

$\angle ROP = 2 \times \angle RSP$ (angle at centre

double angle at circum, same arc

$$\angle RSP = \frac{1}{2}(180 - \alpha)$$

$$= 90 - \frac{\alpha}{2}$$

$\angle RQP$ is cyclic quad

$$\therefore \angle RQP + \angle RSP = 180$$

$$\angle RQP = 180 - (90 - \frac{\alpha}{2})$$

$$= 90 + \frac{\alpha}{2}$$

C

(9)

$$t=0 \quad T=50$$

$$50 = 20 + A$$

$$A = 30$$

$$t=5, \quad T=35$$

$$35 = 20 + 30e^{-5k}$$

$$e^{-5k} = \frac{15}{30}$$

$$k = -\frac{1}{5} \log_e \frac{1}{2}$$

$$= -\frac{1}{5} (\log_e 1 - \log_e 2)$$

$$= \frac{\log_e 2}{5} \quad \text{or} \quad \frac{1}{5} \log_e 2$$

A

(10)

$$\text{For } y = \frac{2}{\pi} \cos^{-1}(2x-1)$$

$$\text{Range } 0 \leq y \leq \pi$$

$$\text{Domain } -1 \leq 2x-1 \leq 1$$

$$0 \leq x \leq 1$$

C

MC B D D A B B D C A C

MATHEMATICS Extension | : Question. 11....

Suggested Solutions

Marks

Marker's Comments

a) $y = e^{2x} \log_e 3x$

$$y' = 2e^{2x} \log_e 3x + \frac{e^{2x}}{x}$$

①

①

b) $\int \frac{1}{1+9x^2} dx$

① for getting $\tan^{-1}$

$$= \frac{1}{3} \int \frac{3}{1+9x^2} dx$$

① The rest of the function

$$= \frac{1}{3} \tan^{-1}(3x) + C$$

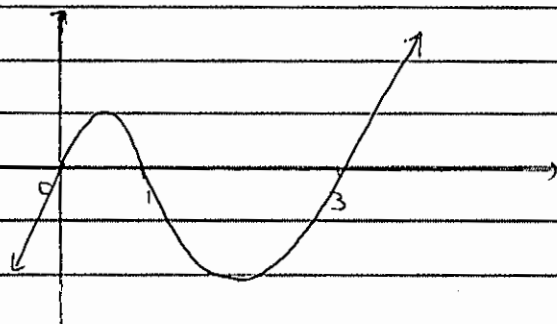
c) $\frac{x^2+3}{x} > 4$

$$x(x^2+3) > 4x^2$$

$$x(x^2-4x+3) > 0$$

$$x(x-1)(x-3) > 0$$

①



$$\therefore 0 < x < 1 \text{ or } x > 3$$

①

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

$$d) \text{ Solve } f(x) = \cos 2x - e^{-x} = 0$$

$$f'(x) = e^{-2x} - 2\sin 2x$$

$$x_2 = 0.4 - \frac{f(0.4)}{f'(0.4)}$$

$$= 0.43 \text{ (2 sig fig)}$$

①

No marks were deducted for
① significant figures.

$$e) \cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$= 1 - 2\sin^2 \theta$$

①

$$\therefore \cos 36 = 1 - 2\sin^2 18$$

$$= 1 - 2\left(\frac{\sqrt{5}-1}{4}\right)^2$$

$$= 1 - 2\left(\frac{5-2\sqrt{5}+1}{16}\right)$$

①

$$= 1 - \left(\frac{6-2\sqrt{5}}{8}\right)$$

$$= \frac{8-6+2\sqrt{5}}{8}$$

$$= \frac{2+2\sqrt{5}}{8}$$

①

$$= \frac{1+\sqrt{5}}{4}$$

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

f)

$$ii) x = 3 - 2\cos 2t$$

$$\dot{x} = 4\sin 2t$$

$$\ddot{x} = 8\cos 2t$$

$$= -4(-2\cos 2t)$$

$$= -4(x-3)$$

$$= -2^2(x-3)$$

which is of the form $\ddot{x} = -n^2(x-b)$

$\therefore$ the motion is simple harmonic

$$iii) \dot{x} = 4\sin 2t$$

$$\text{Max of } \sin 2t = 1$$

$$\therefore \text{Max velocity} = 4 \text{ m/s.}$$

or

max speed when $x = 3$ (centre of motion)

$$\therefore \cos 2t = 0$$

$$t = \frac{\pi}{4}$$

$$\therefore v = 4\sin \frac{\pi}{2} = 4 \text{ m/s}$$

Note: SHM is defined to be a motion such that acceleration is proportional to its displacement from a fixed point but in the opposite direction.

(i)

(i)

(i)

(i)

(i)

MATHEMATICS Extension 1 : Question 12..

Suggested Solutions	Marks	Marker's Comments
a) $\sin 3x = -\frac{1}{2}$ $3x = n\pi + (-1)^n \sin^{-1}(-\frac{1}{2}), n \in \mathbb{Z}$ $3x = n\pi + (-1)^n (-\frac{\pi}{6})$ $\therefore x = n\frac{\pi}{3} + (-1)^n (-\frac{\pi}{6})$ $\therefore A = \frac{\pi}{3}, B = -\frac{\pi}{6}$	1	
b) i) Area of sector = $\frac{1}{2}r^2\theta$ $\therefore \frac{1}{2}r^2\theta = 100$ $r^2\theta = 200$ $\theta = \frac{200}{r^2}$	1	
ii) $\frac{d\theta}{dt} = \frac{d\theta}{dr} \times \frac{dr}{dt}$ $\frac{dr}{dt} = 0.5 \text{ cm s}^{-1}$ $\theta = 200r^{-2}$ $\frac{d\theta}{dr} = -400r^{-3}$ $= -\frac{400}{r^3}$ $\therefore \frac{d\theta}{dt} = -\frac{400}{r^3} \times 0.5$ when $r = 10$ $\frac{d\theta}{dt} = -\frac{400}{10^3} \times 0.5$ $= -0.2$ $\therefore \angle AOB$ is decreasing at a rate of 0.2 radians per second.	1	
c) $(3x - \frac{2}{x})^9$ $T_{R+1} = {}^9C_R (3x)^{9-R} (-\frac{2}{x})^R$ $= {}^9C_R 3^{9-R} (-2)^R x^{9-R-R}$ for coefficient of $x^3: 9-2R=3$ $2R=6$	1	

MATHEMATICS Extension 1 : Question 12...

Suggested Solutions

Marks

Marker's Comments

$$\therefore k=3$$

$$\therefore T_4 = {}^9C_3 3^{9-3} (-2)^3 x^3$$

$$= -489888 x^3$$

$\therefore$ coefficient of x^3 is -489888

$$d)i) P(\text{Thu wins by 3 to 2}) = {}^5C_2 (0.6)^3 (0.4)^2$$

$$= 0.3456$$

$$= 0.346 \text{ (3dp)}$$

ii) Thu must win last game.

Ordering first four games =

$${}^4C_2 (0.6)^2 (0.4)^2$$

last game: 0.6

$$\therefore P = {}^4C_2 (0.6)^2 (0.4)^2 (0.6)$$

$$e)i) \ddot{x} = \frac{-80000}{x^2}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -80000 x^{-2}$$

$$\int d\left(\frac{1}{2} v^2\right) = \int -80000 x^{-2} dx$$

$$\frac{1}{2} v^2 = \frac{80000}{x} + C$$

$$\text{when } v=5, x=6400$$

$$\therefore \frac{1}{2} (5)^2 = \frac{80000}{6400} + C$$

$$\frac{25}{2} = \frac{25}{2} + C$$

$$\therefore C=0$$

$$\therefore v^2 = \frac{160000}{x}$$

rocket comes to rest when $v=0$

$$0 = \frac{160000}{x}$$

$\therefore$ no solution

$\therefore$ rocket does not come to rest.

1

1

1

1

1

2

1 for explanation

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
ii) $x = 6400 + 3600$ $= 10000$ $\therefore v^2 = \frac{160000}{10000}$ $= 16$ $v = \pm 4 \text{ km/s}$ Since initial velocity is positive and rocket never comes to rest ie. it will not change direction $\therefore v > 0$ $\therefore v = 4 \text{ km/s}$		$\Rightarrow$ must have explanation to why $v > 0$.

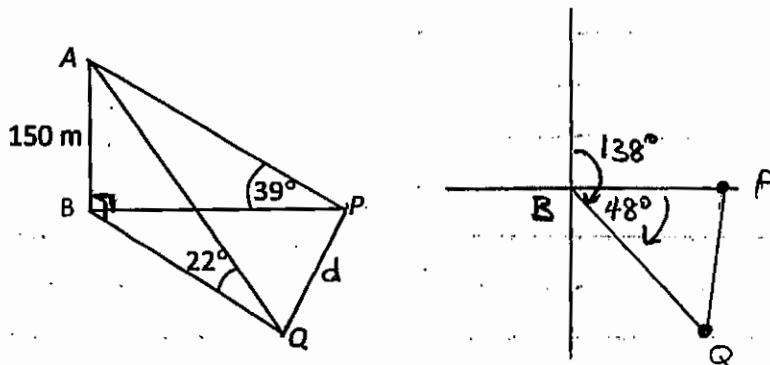
MATHEMATICS Ext 1: Question: 13

Suggested Solutions

Marks

Marker's Comments

(a)



$$\angle PBQ = 138^\circ - 90^\circ = 48^\circ$$

Letting $d = PQ$, then, by cosine rule,

$$d^2 = BQ^2 + BP^2 - 2(BQ)(BP)\cos 48^\circ$$

$$\text{Now, } \tan 39^\circ = \frac{150}{BP} \Rightarrow BP = 150 \cot 39^\circ$$

and

$$\tan 22^\circ = \frac{150}{BQ} \Rightarrow BQ = 150 \cot 22^\circ$$

So

$$d^2 = (150 \cot 39^\circ)^2 + (150 \cot 22^\circ)^2$$

$$- 2(150)^2 \cot 39^\circ \cot 22^\circ$$

$$= 80114.85 \dots \text{ m}^2$$

So

$$PQ = \sqrt{d^2} = 283.05 \dots \text{ m}$$

$$= 280 \text{ m}$$

to nearest 10 m

Handled well,
Minor issues
such as
failing to take
 $\sqrt{\quad}$ or
mistake with
signs in
cosine rule.

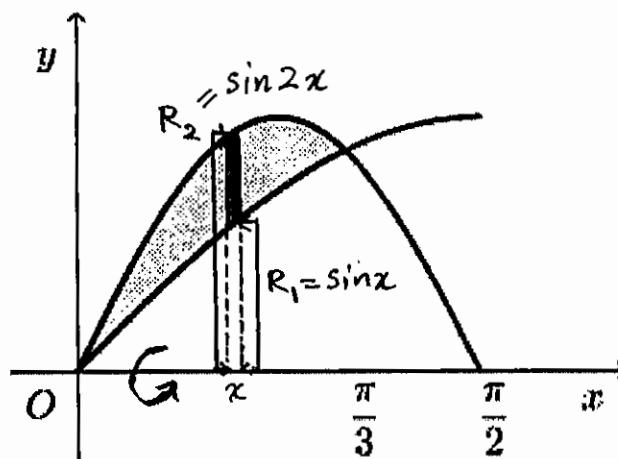
MATHEMATICS Ext 1: Question: 13

Suggested Solutions

Marks

Marker's Comments

(b) ✓.



(volume) = (larger cylinder)
- (smaller cylinder)

$$= \pi R_2^2 h - \pi R_1^2 h$$

$$= \pi (R_2^2 - R_1^2) h$$

Where, at x , $R_2 = \sin 2x$, $R_1 = \sin x$
and $h = \delta x$.

Also, curves intersect where

$$\sin 2x = \sin x$$

$$2 \sin x \cos x = \sin x$$

So, either $\sin x = 0$ or $\cos x = 1/2$.

Over domain: $x = 0$ or $x = \pi/3$.

Primary error
in setup:

taking

$$(R_2 - R_1)^2$$

instead of

$$R_2^2 - R_1^2$$

- you're
subtracting
one cylindrical
volume from
another.

MATHEMATICS Ext 1: Question: 13

Suggested Solutions

Marks

Marker's Comments

$$\text{So, } V = \pi \int_0^{\pi/3} \sin^2 2x - \sin^2 x \, dx$$

$$= \pi \int_0^{\pi/3} \frac{1 - \cos 4x}{2} - \frac{1 - \cos 2x}{2} \, dx$$

$$= \frac{\pi}{2} \int_0^{\pi/3} \cos 2x - \cos 4x \, dx$$

$$= \frac{\pi}{2} \left[\frac{1}{2} \sin 2x - \frac{1}{4} \sin 4x \right]_0^{\pi/3}$$

$$= \frac{\pi}{2} \left[\left(\frac{1}{2} \sin \frac{2\pi}{3} - \frac{1}{4} \sin \frac{4\pi}{3} \right) - 0 \right]$$

$$= \frac{\pi}{2} \left[\frac{1}{2} \cdot \frac{\sqrt{3}}{2} - \frac{1}{4} \left(-\frac{\sqrt{3}}{2} \right) \right]$$

$$= \frac{\pi}{2} \cdot \frac{3\sqrt{3}}{8}$$

$$= \frac{3\pi\sqrt{3}}{16} \text{ units}^3$$

1

Factor of π missing from several responses; HSC will deduct a mark for this, so pay attention to your initial setup.

1

Another issue, integrand ordered incorrectly (e.g. $\sin^2 x - \sin^2 2x$), leading to negative volume $\rightarrow$ shouldn't happen if you're integrating correctly, so check your work. May admit to error and 'drop' sign.

1

MATHEMATICS Ext 1: Question: 13

Suggested Solutions

Marks

Marker's Comments

(c) Let $S(n)$ be statement

$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \dots + \frac{1}{n(n+1)(n+2)} = \frac{1}{4} - \frac{1}{2(n+1)(n+2)} \quad \text{well.}$$

where $n \in \mathbb{N}$, $n \geq 1$.

(I) Show $S(1)$ true:

$$\text{LHS} = \frac{1}{1 \times (1+1)(1+2)} \quad \text{RHS} = \frac{1}{4} - \frac{1}{2(1+1)(1+2)}$$

$$= \frac{1}{6}$$

$$= \frac{1}{4} - \frac{1}{12}$$

$$= \frac{1}{6}$$

$\therefore \text{LHS} = \text{RHS}$, so $S(1)$ true.

(II) Assume true for $n=k$; i.e. assume $S(k)$ true where $k \in \mathbb{N}$.

Then

$$\frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{k(k+1)(k+2)} = \frac{1}{4} - \frac{1}{2(k+1)(k+2)} \quad \text{--- (*)}$$

is taken to be the case.

For $S(k+1)$:

$$\left(\frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{k(k+1)(k+2)} \right) + \frac{1}{(k+1)((k+1)+1)((k+1)+2)}$$

$$= \frac{1}{4} - \frac{1}{2(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

by hypothesis, (*)

(1)

Handled

Only issue with some:

need to finish with a proper concluding statement.

MATHEMATICS Ext 1: Question: 13

Suggested Solutions

Marks

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$$= \frac{1}{4} - \left[\frac{1}{2(k+1)(k+2)} \cdot \frac{(k+3)}{(k+3)} - \frac{2}{2(k+1)(k+2)(k+3)} \right]$$

$$= \frac{1}{4} - \left[\frac{(k+1)}{2(k+1)(k+2)(k+3)} \right]$$

$$= \frac{1}{4} - \frac{1}{2((k+1)+1)((k+1)+2)}$$

①

which is of the required form.

Hence we've shown that if

$S(k)$ is true, then $S(k+1)$ is

true: $S(k) \rightarrow S(k+1)$ — ②

(III) Hence by principle of mathematical

induction, since $S(1)$ is true,

it follows by ② that $S(2)$ is

true; since $S(2)$ true, $S(3)$

is true, etc.

Hence $S(n)$ is true for all

integers $n \geq 1$ //

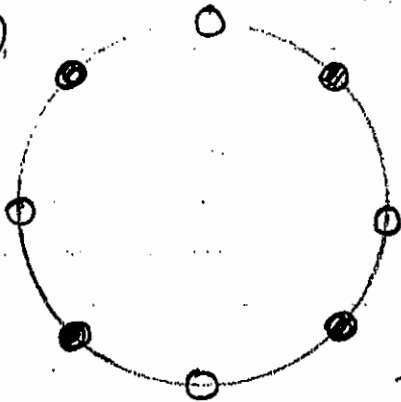
MATHEMATICS Ext 1: Question: 13

Suggested Solutions

Marks

Marker's Comments

(d)



arrange all
males, say,
giving

$(4-1)! = 3!$
arrangements.

Then first F
partner can be seated 2
ways. Once this occurs,
all other F partners have no
choice but to sit in a
determined position.

So, $3! \times 2 = 12$ ways.

1

Not handled
well.

First mark
for 'reasonable'
start.

1

Final mark
for correct
answer.

MATHEMATICS Ext 1: Question: 13

ELSE

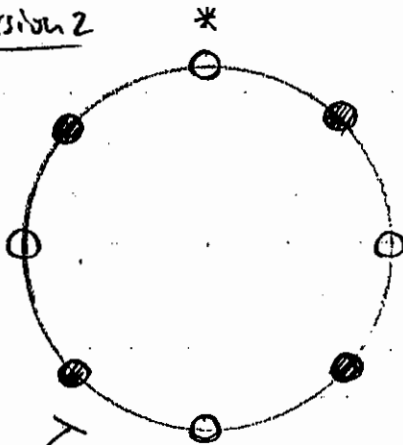
Suggested Solutions

Marks

Marker's Comments

version 2

(d)



Let couples be assigned as

$$\{M_i, F_i\}$$

where $i = 1, 2, 3, 4$.

whog,
place
 M_1 at
*

$$\frac{M_1}{*} \text{ --- } \frac{F_1}{*}$$

OR

$$\frac{M_1}{*} \frac{F_1}{*} \text{ --- } \text{ --- } \text{ --- } \text{ --- } \text{ --- } \text{ --- } \text{ --- } \text{ ---}$$

Then F_1 must be placed at posⁿ ② or ⑧ (depending on which side of M_1 she sits).

Choices for male/female couples are then fixed as

$$\frac{1}{M_1} \times \frac{3}{F_1} \times \frac{1}{M_2} \times \frac{2}{F_2} \times \frac{1}{M_3} \times \frac{1}{F_3} \times \frac{1}{M_4} \times \frac{1}{F_4}$$

OR

$$\frac{1}{M_1} \times \frac{1}{F_1} \times \frac{3}{M_2} \times \frac{1}{F_2} \times \frac{2}{M_3} \times \frac{1}{F_3} \times \frac{1}{M_4} \times \frac{1}{F_4}$$

So, $3! \times 2 = 12$ ways.

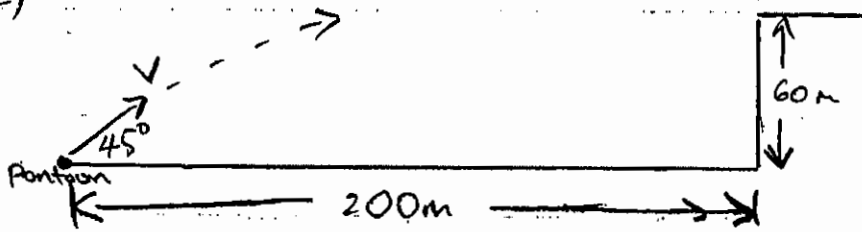
MATHEMATICS Ext 1: Question: 13

Suggested Solutions

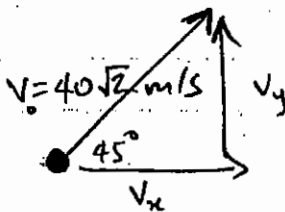
Marks

Marker's Comments

(e)



(i)



Taking $g = 10 \text{ m s}^{-2}$:

$$v_x = 40\sqrt{2} \sin 45^\circ = 40 \text{ m/s}$$

$$v_y = 40\sqrt{2} \cos 45^\circ = 40 \text{ m/s}$$

Horizontal trajectory:

$$\ddot{x} = 0$$

$$\dot{x} = c_1 \text{ for all } t.$$

$$\text{Hence } \dot{x}(0) = c_1 = 40 \text{ m/s:}$$

$$\text{so } \dot{x} = 40 \quad (\text{hold units})$$

$$\text{Then } x = 40t + c_2$$

Part (i)
handled well.

Only small
issues:

* not evaluating
 $\sin 45^\circ, \cos 45^\circ$,
or doing so
incorrectly

* not substituting
values given
for pronumerals

MATHEMATICS Ext 1: Question: 13

Suggested Solutions

Marks

Marker's Comments

Now, $x(0) = 0$, so $c_2 = 0$,

so $\boxed{x(t) = 40t}$ — ①

Vertical trajectory:

$\ddot{y} = -10$ (hold units)

so $\dot{y} = -10t + c_3$

Now, $\dot{y}(0) = 40 \text{ m/s}$, so

$c_3 = 40 \text{ m/s}$.

$\therefore \dot{y} = -10t + 40$

Hence

$$y = -\frac{10t^2}{2} + 40t + c_4$$

$$= -5t^2 + 40t + c_4$$

Now, $y(0) = 0$, so $c_4 = 0$.

Hence

$\boxed{y = -5t^2 + 40t}$

1

We look first for x when $y = 60\text{m}$ (height of cliff), expecting this type of situation



Then, by (i)

$$60 = -5t^2 + 40t$$

$$\Rightarrow t^2 - 8t + 12 = 0$$

$$\Rightarrow (t-2)(t-6) = 0$$

When $t = 2$, object reaches 60m in ascent; $t = 6$, reaches 60m in descent; take $t = 6\text{s}$.

1 First mark for 'logical' start.

When $x(6) = 40(6)$
 $= 240\text{m}.$

$$240 - 200 = 40$$

The rocket lands 40m from edge of cliff

1

①

MATHEMATICS Extension 1: Question 14

Suggested Solutions	Marks Awarded	Marker's Comments
14 a) $y = \frac{1}{\sqrt{1+x^2}} \quad x \leq 0$ $x = \frac{1}{\sqrt{1+y^2}} \quad y < 0$ $x^2 = \frac{1}{1+y^2}$ $1+y^2 = \frac{1}{x^2}$ $y^2 = \frac{1}{x^2} - 1$ $y = \pm \sqrt{\frac{1}{x^2} - 1}$ since $y < 0$ $y = -\sqrt{\frac{1}{x^2} - 1}$ b) $x^2 = 8y$ $y = \frac{x^2}{8}$ $\frac{dy}{dx} = \frac{2x}{8}$ $= \frac{x}{4}$ at $x = 4p$ $\frac{dy}{dx} = p$ $\therefore y - 2p^2 = p(x - 4p)$ $y = px - 4p^2 + 2p^2$ $y = px - 2p^2$	1 1	well done for correct answer students need to think about domain given.

2

MATHEMATICS Extension 1 : Question 14

Suggested Solutions	Marks Awarded	Marker's Comments
Similarly $y = qx - 2q^2$ $\therefore y = px - 2p^2$ $\quad y = qx - 2q^2$ $0 = (p - q)x - 2(p^2 - q^2)$ $(p - q)x = 2(p - q)(p + q)$ $x = 2(p + q)$ $y = 2q(p + q) - 2q^2$ $\quad = 2pq + 2q^2 - 2q^2$ $\quad = \underline{2pq}$ $M(2(p + q), 2pq)$ (ii) $\tan 45^\circ = \left \frac{n_1 - n_2}{1 + n_1 n_2} \right$ $n_p = p \quad n_q = q$ $1 = \frac{ p - q }{ 1 + pq }$ $1 + pq = p - q$ e (ii) $p - q = 1 + pq$ $\therefore (p - q)^2 = (1 + pq)^2 \quad \text{--- (1)}$ $x = 2(p + q) \quad y = 2pq \quad \text{--- (2)}$	1 1 1	question said find students needed to derive formulae for tangents

Suggested Solutions	Marks Awarded	Marker's Comments
MATHEMATICS Extension 1: Question 14...		
$p^2 - 2pq + q^2 = 1 + 2pq + p^2q^2$ Now $p+q = \frac{x}{2}$ $(p+q)^2 = p^2 + q^2 + 2pq$ $\frac{x^2}{4} = p^2 + q^2 + y$ Since $y = 2pq$ $\therefore p^2 + q^2 - 2pq = \frac{x^2}{4} - y - y$ $1 + 2pq + p^2q^2 = 1 + y + \frac{y^2}{4}$ $\therefore \frac{x^2}{4} - 2y = 1 + y + \frac{y^2}{4}$ $x^2 - 8y = 4 + 4y + y^2$ $x^2 - y^2 - 12y - 4 = 0$ (c) let $u = t^2 + 2$ $\frac{du}{dt} = 2t$ $\int t^2 \sqrt{t^2 + 2} \, t \, dt$ $= \int (u-2) \sqrt{u} \frac{du}{2}$ $= \frac{1}{2} \int u^{3/2} - 2u^{1/2} \, du$	1 1	Needed to explain what you were doing students did not put brackets in or had $\sqrt{u-2}$

(4)

Suggested Solutions	Marks Awarded	Marker's Comments
MATHEMATICS Extension 1 14: Question.....		
$= \frac{1}{2} \left(2u^{5/2} - \frac{4u^{3/2}}{3} \right)$ $= \frac{1}{5} (t^2+1)^{5/2} - \frac{2(t^2+1)^{3/2}}{3}$ d) $f(x) = 2\cos^{-1}x - 1 \leq x \leq 1$ $g(x) = \sin^{-1}(2x^2-1) - 1 \leq x \leq 0$ (c) $y = f(x)$ (u) $f(x) = \frac{-2}{\sqrt{1-x^2}}$ $g(x) = \frac{1}{\sqrt{1-(2x-1)^2}} \cdot 4x$ $= \frac{4x}{\sqrt{1-4x^2+4x^2-1}}$ $= \frac{4x}{2 x \sqrt{1-x^2}}$ Now if $-1 < x < 0$ then $x = -x$ so $= \frac{4x}{-2x\sqrt{1-x^2}}$	1 1 1	Had to integrate substitute back if you lost fractional index did not get final mark CFE. well done but think about scale $\pi = 3.14$ same graphs very distorted. Needn't + clear explanation not sufficient just too little $-1 < x < 0$.

MATHEMATICS Extension 1: Question 14

Suggested Solutions

Marks Awarded

Marker's Comments

$$\therefore g'(x) = \frac{-2}{2\sqrt{1-x^2}}$$

$$\therefore f'(x) = g'(x)$$

$$\text{ii) } \int f'(x) = g'(x)$$

$$\text{then } f(x) = g(x) + C$$

$$\sin^{-1}(2x^2-1) = 2\cos^{-1}x + C$$

when $x=0$

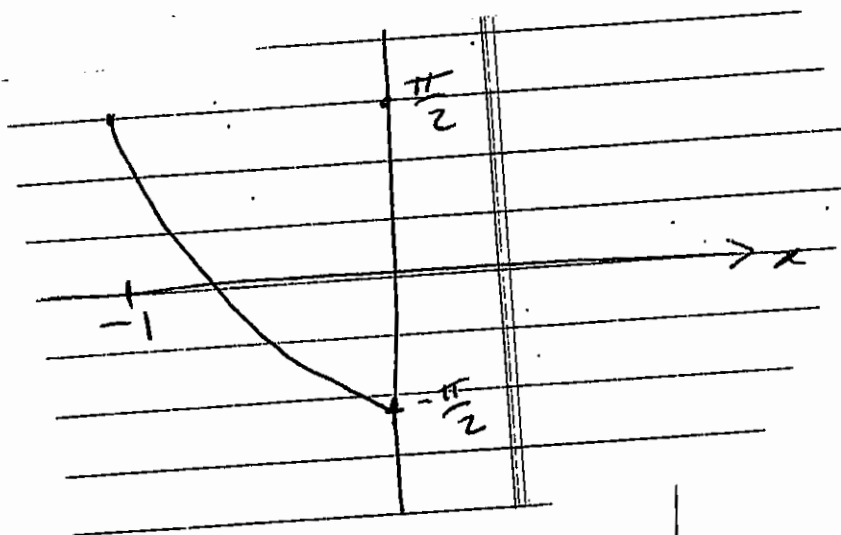
$$\sin^{-1}(-1) = 2\cos^{-1}0 + C$$

$$-\frac{\pi}{2} = \pi + C$$

$$C = -\frac{3\pi}{2}$$

$$\therefore g(x) = f(x) - \frac{3\pi}{2}$$

(iv)



value had
to be in
domain
 $-1 \leq x \leq 0$

some, hole is
didn't write
 $g(x)$ in terms
of $f(x)$.

MATHEMATICS Extension 1: Question 14

Suggested Solutions	Marks Awarded	Marker's Comments
$\therefore g'(x) = \frac{-2}{2\sqrt{1-x^2}}$ $\therefore f'(x) = g'(x)$ iii) If $f'(x) = g'(x)$ then $f(x) = g(x) + C$ $\sin^{-1}(2x^2-1) = 2\cos^{-1}x + C$ when $x=0$ $\sin^{-1}(-1) = 2\cos^{-1}0 + C$ $-\frac{\pi}{2} = \pi + C$ $C = -\frac{3\pi}{2}$ $\therefore g(x) = f(x) - \frac{3\pi}{2}$	1 1	value had to be in domain $-1 \leq x \leq 0$ some, kids did not write $g(x)$ in terms of $f(x)$.
(iv) 